

To enable the generation of P-EC, it is essential to formally express the violation patterns of the four defining properties of *strict weak ordering* —**ir-reflexivity**, **asymmetry**, **transitivity**, and **consistency**.

To simplify these expressions, we introduce a prioritization among the properties: **ir-reflexivity** > **asymmetry** > **transitivity** > **consistency**. When detecting a violation of a given property, we assume that all higher-priority properties hold. This assumption reduces unnecessary corner cases and simplifies the formulation of each violation. Additionally, for comparators returning `int`, We also eliminate corner cases where only one side is nonzero when violating **ir-reflexivity**.

The violation patterns of the first three properties can be expressed as follows:

- For the return type `bool`:

- **Ir-reflexivity violation:**

$$\text{cmp}(a, a) = \text{true} \quad (1)$$

- **Asymmetry violation:**

$$\text{cmp}(a, b) = \text{true} \wedge \text{cmp}(b, a) = \text{true} \quad (2)$$

- **Transitivity violation:**

$$\begin{aligned} \text{cmp}(a, b) = \text{true} \wedge \text{cmp}(b, c) = \text{true} \\ \wedge \text{cmp}(c, a) = \text{true} \end{aligned} \quad (3)$$

- For the return type `int`:

- **Ir-reflexivity violation:**

$$\text{cmp}(a, a) \neq 0 \quad (4)$$

- **Asymmetry violation:**

$$\begin{aligned} (\text{cmp}(a, b) > 0 \wedge \text{cmp}(b, a) > 0) \\ \vee (\text{cmp}(a, b) < 0 \wedge \text{cmp}(b, a) < 0) \end{aligned} \quad (5)$$

- **Transitivity violation:**

$$\begin{aligned} \text{cmp}(a, b) < 0 \wedge \text{cmp}(b, c) < 0 \\ \wedge \text{cmp}(c, a) < 0 \end{aligned} \quad (6)$$

*Note:* When checking violations of **transitivity** in the case where  $\text{cmp}(a, b) > 0$ , the comparator satisfies **asymmetry** because of the priority. So the violating elements can be reversed to yield an equivalent violation for the  $\text{cmp}(a, b) < 0$  case. Therefore, it is sufficient to consider only one direction of the violation.

Finally, we formalize the violation of **consistency**, which differs from other properties in that it inherently involves equality relation.

For comparators returning `int`, There are two cases, condition (7) and condition (8). For the condition (7), we can assume  $\text{cmp}(a, c) < 0$ . So the condition (7) and the condition (8) can both derive the condition (9). Then, because the comparator satisfies **transitivity**, it is impossible for both  $\text{cmp}(a, b)$  and  $\text{cmp}(b, c)$  to be strictly greater than zero at the same time in the condition (9). Therefore, we can safely assume  $\text{cmp}(a, b) = 0$  without loss of generality. This deduction allows us to conclude that the condition (9) represents the union of the condition (7) and the condition (8). So the condition (9) is both a necessary and sufficient condition for detecting violations of **consistency**.

For comparators that return `bool`, two elements `a` and `b` are considered equal if and only if  $\text{cmp}(a, b) = \text{false} \wedge \text{cmp}(b, a) = \text{false}$ . A violation of **consistency** occurs when `a` and `b` are equal but produce different comparison results with a third element `c`. Without loss of generality, We can suppose  $\text{cmp}(a, b) = \text{cmp}(b, a) = \text{false}$ ,  $\text{cmp}(b, c) = \text{false}$ ,  $\text{cmp}(a, c) = \text{true}$ , so the comparator also violates the **transitivity** of the false relation. On the other hand, if the comparator violates the **transitivity** of the false relation, We can suppose  $\text{cmp}(a, b) = \text{false}$ ,  $\text{cmp}(b, c) = \text{false}$ ,  $\text{cmp}(a, c) = \text{true}$ . If  $\text{cmp}(b, a) = \text{true}$ , then according to the **transitivity** and  $\text{cmp}(a, c) = \text{true}$ , we would have  $\text{cmp}(b, c) = \text{true}$ , this is contradictory. So  $\text{cmp}(b, a) = \text{false}$ , that is a violation of **consistency**. Therefore, the violation of **transitivity** of the false relation is equivalent to the violation of **consistency**.

$$\text{cmp}(a, b) = 0 \wedge \text{cmp}(b, c) = 0 \wedge \text{cmp}(a, c) \neq 0 \quad (7)$$

$$\text{cmp}(a, b) = 0 \wedge \text{cmp}(b, c) > 0 \wedge \text{cmp}(a, c) < 0 \quad (8)$$

$$\text{cmp}(a, b) \geq 0 \wedge \text{cmp}(b, c) \geq 0 \wedge \text{cmp}(a, c) < 0 \quad (9)$$